KAPIL ANANDH P

Generative AI | AI/ML Engineer | Python | LangGraph | LangChain | LLMs | NLP | AWS

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Portfolio : https://kapilanandh.github.io/

PROFESSIONAL SUMMARY

Generative AI and AI/ML Engineer with 1.5+ years of hands-on experience building LLM-based applications and AI agents in Python using LangGraph, LangChain and cloud AI platforms (AWS Bedrock, SageMaker, GCP). Strong in Python, algorithms, deep learning, NLP, image segmentation and processing, and data collection and data processing for model training, evaluation and deployment. Fine-tuned LLMs (LoRA/QLoRA) to 94% accuracy with 80% cost reduction and cut inference latency by 25%. Skilled at translating business requirements into technical tasks, documenting AI/ML projects, and solving problems in a collaborative, fast-paced environment that values continuous learning and innovation.

TECHNICAL SKILLS

Programming: Python (primary), SQL, TypeScript, Node.js, Bash, Object-Oriented Programming, Algorithms, FastAPI, Flask, Django, REST APIs, WebSockets

Generative AI & LLMs: Large Language Models (LLM), LLM Fine-Tuning (LoRA, QLoRA), RAG, Prompt Engineering, Function Calling, NL2SQL, RLHF, Embedding Models, Hugging Face, Anthropic Claude, Gemini, Azure OpenAI, MedGemma

AI Agents: LangGraph, LangChain, LlamaIndex, Multi-Agent Systems, Agentic Workflows, Tool Calling, Amazon Bedrock AgentCore, Mastra, Semantic Kernel, Temporal

ML, Deep Learning & NLP: Machine Learning, Deep Learning, NLP, Transformer Architecture, TensorFlow, PyTorch, Keras, Scikit-learn, Time-Series Forecasting (ARIMA), Model Evaluation

Computer Vision & Speech: Image Segmentation & Processing, OpenCV, Vision Transformers (ViT), OCR, Amazon Textract, Amazon Rekognition, Whisper (STT), F5-TTS

Data Processing & Databases: Data Collection, Data Preprocessing, Data Processing, ETL Pipelines, Feature Engineering, Vector Databases (Pinecone, ChromaDB, FAISS, Weaviate, OpenSearch Serverless), PostgreSQL, MySQL, Redis, DynamoDB

Cloud & MLOps: AWS (Bedrock, SageMaker, Lambda, EC2, S3, Glue, Athena, CloudWatch, CDK, Step Functions, EventBridge), GCP, Docker, Terraform, CI/CD (GitHub Actions, GitLab), Git

Inference Optimization: Model Quantization, ONNX Runtime, vLLM, Flash Attention, GPU Optimization, Latency Optimization, Scalability Testing

Governance: Responsible AI, Data Privacy, ISO 27001, Bedrock Guardrails

Professional Skills: Problem Solving, Good Communication, Collaboration, Mentoring, Documentation, Reliability, Innovation

PROFESSIONAL EXPERIENCE

Associate AI/ML Engineer | OneData Software Solutions Private Limited, Coimbatore **12/2025 – Present**

- Fine-tuned MedGemma LLMs (LoRA/QLoRA) in Python on 50K+ clinical records, handling data collection and preprocessing for training and evaluation; achieved 94% accuracy and 80% cost reduction through on-prem deployment.
- Architected an end-to-end multimodal AI platform (LLM, OCR, computer vision, speech, RAG) from development to production, serving 10+ hospitals under ISO 27001 data-privacy compliance.
- Built multi-agent GenAI systems and agentic workflows with an RLHF feedback loop for diagnosis and prescription suggestions; developed and maintained MLOps pipelines on AWS (Bedrock, SageMaker, Lambda) and on-prem GPU infrastructure.
- Fine-tuned OCR models to 94% accuracy, digitizing 30+ years of hospital records; engineered a speech-to-text (STT) system for medical transcription and automatic translation.
- Collaborated with clients, stakeholders and cross-functional teams to translate business requirements into technical tasks; documented solution architectures and mentored junior engineers.

AI/ML Engineer | Charky Labs Pvt. Ltd., Coimbatore **04/2025 – 11/2025**

- Built a real-time speech-to-speech platform in Python (Whisper STT, F5-TTS voice cloning) with sub-second inference latency.
- Optimized inference pipelines using ONNX Runtime, vLLM and Flash Attention on GPU for performance, scalability and reliability; benchmarked 30+ open-source models across speech and NLP algorithms.
- Automated AWS CI/CD deployment, cutting inference latency by 25%.

KEY PROJECTS

OneReach v2 – Agentic AI | Sophia, autonomous AI sales agent **09/2026 – Present**

- Building an AI agent on the Mastra framework with provider-agnostic LLMs (OpenAI, Anthropic, Gemini): 78 tools across 35 capabilities with semantic tool retrieval to keep prompts within token limits; durable agent workflows on Temporal.
- Built streaming AI chat (CopilotKit/AG-UI, SSE) and multi-format document ingestion (PDF, DOCX, PPTX, image OCR) with citation-backed RAG search; full-stack delivery on NestJS, PostgreSQL and Next.js.

OneLingo – Real-Time : Translation Platform **09/2026 – Present**

- Building a multi-tenant desktop and web platform streaming live captions and translation over WebSockets with selectable STT/LLM engines ; Node.js/PostgreSQL backend with role-based access, audit logs and usage analytics.

Blacksmith Research – Fashion Trends Analytics **04/2026 – 08/2026**

- Engineered an AI trend-detection pipeline using Amazon Bedrock (Nova Pro) vision models on catalog images from 8 fashion portals, stored in a data lake (S3, Glue, Athena); built a conversational analytics agent for natural-language trend queries.

HelPiere – Multi-Agent Digital Concierge **05/2026 – 07/2026**

- Designed a multi-agent GenAI concierge using LangGraph on Amazon Bedrock AgentCore, with an orchestrator routing between accommodation and food-delivery agents; integrated Expedia and UberEats APIs behind a bilingual, JWT-secured REST API for WhatsApp Business.

Abaton – Horus MLOps (SageMaker Forecasting Pipeline) **05/2026 – 06/2026**

- Trained an ARIMA forecasting model on Amazon SageMaker with Step Functions (95% confidence-interval accuracy); automated the full ML lifecycle (data extraction, scheduled training/inference, prediction persistence) using AWS CDK and EventBridge.

Stratagraph Energy Group – AWS & GenAI Platform Modernization **03/2026 – 06/2026**

- Built an AI Log Quality Grading system with Amazon Textract, Rekognition and Bedrock Claude for near-real-time grading of mudlog reports; created an ~8TB data lake with Titan embeddings in OpenSearch Serverless powering a RAG search assistant with citation-backed answers (Terraform IaC, CI/CD, ISO 27001).

Perzona – Agentic AI & NL2SQL Analytics **02/2026 – 04/2026**

- Architected a multi-tenant NL2SQL conversational analytics platform on Amazon Bedrock (Claude Sonnet 4.5), Lambda, CDK and DynamoDB serving 40+ client tenants; implemented three-layer security (Bedrock Guardrails + SELECT-only SQL validator) with CloudWatch monitoring and audit logging.

EDUCATION

Master of Technology (M.Tech), Data Science | Kumaraguru College of Technology, Coimbatore **09/2023 – 04/2025**

CERTIFICATIONS

- IELTS – Overall Band Score 7.5
- AWS Certified Machine Learning Engineer – Associate – Amazon Web Services
- TensorFlow Developer Certificate – TensorFlow
- Claude Partner Badge (Claude Code) – Anthropic
- AI Fluency for Builders – CodePath × Anthropic